HR ANALYTICS

Employee Attrition Prevention Report

A data-driven analysis of employee resignation patterns
using Exploratory Data Analysis and Machine Learning
to identify root causes and prevent future attrition.

1,470 Employees Analysed	**16.12% Attrition Rate**	**88.1% Model Accuracy**	**6 Key Recommendations**

Prepared by
Ankita Daweshar
Data Analyst Intern | June 2026

Python Pandas · Seaborn Scikit-learn	Power BI 3-Page Dashboard KPI Cards · Slicers	Dataset IBM HR Analytics 1,470 Records

1. Introduction

Employee attrition — the voluntary resignation of employees — is one of the most costly and disruptive challenges facing modern organizations. When a skilled employee leaves, the company bears recruitment costs, training expenses, and a significant loss of institutional knowledge. Studies suggest that replacing an employee can cost between 50% to 200% of their annual salary. For large organizations employing thousands of people, even a modest attrition rate of 15-20% can translate into millions of rupees in annual losses.

Traditional HR management has largely been reactive — organizations respond to resignations only after they happen. However, with the rise of data science and machine learning, it is now possible to take a proactive approach. By analyzing historical employee data, HR teams can identify patterns that predict which employees are most likely to resign — and intervene before the resignation actually occurs. This shift from reactive to predictive HR management is one of the most significant transformations in modern workforce strategy.

This report presents the findings of an HR Analytics project conducted on the IBM HR Employee Attrition dataset, comprising 1,470 employee records across 35 attributes including age, department, job role, monthly income, overtime status, job satisfaction, work-life balance, and years since last promotion. The goal was to identify the primary drivers of attrition and provide actionable, data-backed recommendations to help HR teams reduce voluntary employee turnover effectively and sustainably.

The analysis was conducted in six structured phases: dataset setup and exploration, exploratory data analysis (EDA), machine learning model development, Power BI dashboard creation, prevention report generation, and final project documentation. Each phase built upon the previous to deliver a complete, end-to-end HR analytics solution that bridges the gap between raw employee data and strategic HR decision-making.

The IBM HR Analytics dataset was chosen for this project due to its comprehensive coverage of employee attributes, its wide use in the data science community, and its realistic representation of organizational HR challenges. With 35 features spanning personal, professional, financial, and psychological dimensions of employee experience, it provides an ideal foundation for both exploratory analysis and predictive modeling.

The insights derived from this project are not limited to IBM or any single organization. The patterns identified in employee attrition are broadly applicable across industries — from IT and banking to manufacturing and retail. Any organization that collects employee data can adapt the methodology presented here to build their own attrition prediction system and develop targeted employee retention strategies.

Ultimately, the goal of this project is to demonstrate that data science is not just a technical discipline — it is a powerful tool for solving real human problems. By understanding why employees leave, organizations can create better workplaces, improve employee wellbeing, and build stronger, more resilient teams.

2. Abstract

This project applies Exploratory Data Analysis (EDA) and Machine Learning (Random Forest Classifier) to the IBM HR Attrition dataset to understand and predict employee resignations. The dataset contains 1,470 employee records with 35 features covering personal demographics, job characteristics, compensation details, satisfaction scores, and work history. After thorough data cleaning and exploratory analysis, a Random Forest Classifier was trained on an 80:20 train-test split to predict the binary target variable — Attrition (Yes/No).

Key findings reveal that overtime work, low monthly income, young employee age (28-35), low job satisfaction, and prolonged absence of promotions are the strongest predictors of attrition. The Random Forest model achieved 88.1% accuracy, with Monthly Income, OverTime, and Age identified as the top three feature importance predictors. A 3-page interactive Power BI dashboard was developed to provide HR managers with a visual, filterable view of attrition patterns across departments, job roles, satisfaction levels, and demographics.

Based on these data-driven findings, six targeted HR interventions are recommended to address the root causes of attrition: implementing an overtime management policy, conducting a comprehensive salary review, creating a young talent retention program, introducing regular promotion and recognition cycles, strengthening work-life balance initiatives, and deploying a predictive attrition alert system. If adopted, these measures have the potential to significantly reduce the current 16.12% attrition rate and improve overall employee retention.

This project demonstrates the practical application of data science in human resource management, showing how analytical methods can transform reactive HR practices into proactive, evidence-based workforce strategies. All project code, notebooks, Power BI dashboard, and deliverables are version-controlled and available on GitHub.

The entire project was executed using open-source Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn, along with Power BI Desktop for dashboard creation. The methodology is fully reproducible and can be applied to any organization's HR dataset with minimal modifications. The predictive model, once deployed, can serve as an early warning system that flags at-risk employees for proactive HR intervention.

The findings of this project highlight a critical insight: employee attrition is not random. It follows identifiable patterns rooted in compensation, workload, career growth, and job satisfaction. Organizations that recognize and act on these patterns early will have a significant competitive advantage in attracting and retaining top talent in an increasingly competitive job market.

Keywords: HR Analytics, Employee Attrition, Random Forest, Exploratory Data Analysis, Power BI, Machine Learning, Workforce Management, Employee Retention, IBM HR Dataset, Predictive Modeling.

3. Key Findings from Data Analysis

Metric	Value	Insight
Total Employees	1,470	IBM HR Dataset
Employees Who Left	237	16.12% Attrition Rate
Employees Who Stayed	1,233	83.88% Retention Rate
ML Model Accuracy	88.1%	Random Forest Classifier
Top Predictor	Monthly Income	Feature Importance #1
2nd Top Predictor	OverTime	Feature Importance #2
3rd Top Predictor	Age	Feature Importance #3

3.1 Finding 1 — Overtime is the Strongest Behavioral Driver

Out of 237 employees who left, 110 (46.4%) were working overtime — despite overtime workers representing only 28% of the total workforce. This indicates excessive workload is a critical burnout factor.

3.2 Finding 2 — Low Monthly Income Correlates Strongly with Attrition

Employees earning Rs. 2,500 to Rs. 5,000 per month constitute the majority of resignations. Monthly Income was ranked as the #1 most important feature by the Random Forest model.

3.3 Finding 3 — Young Employees (Age 28-35) Leave the Most

Age group 28-35 shows the highest attrition count. This demographic has higher career mobility and is actively exploring better opportunities, making them the most vulnerable group.

3.4 Finding 4 — Job Satisfaction Level 3 Has Highest Attrition

Employees with Job Satisfaction score of 3 (moderately satisfied) show the highest attrition (73 employees). Satisfaction alone is insufficient — growth, recognition, and compensation must all align.

3.5 Finding 5 — Sales Department Has Highest Attrition Risk

Despite fewer employees than R&D;, Sales shows disproportionately high attrition. High targets, frequent travel, and performance pressure contribute to burnout and voluntary exits.

3.6 Finding 6 — Poor Work-Life Balance Drives Resignations

Employees with Work-Life Balance rating of 3 show the highest attrition (127 employees). Combined with overtime, this creates a compounding retention risk for the organization.

4. Attrition Prevention Recommendations

Based on the data analysis findings, the following six recommendations are proposed for the HR team to reduce employee attrition:

Recommendation 1 — Implement an Overtime Management Policy

Introduce a strict overtime monitoring system. Employees should not work more than 10 hours of overtime per week. Introduce compensatory time-off, overtime bonuses, or workload redistribution for teams with consistently high overtime hours. Target the Sales and R&D; departments first, as these show the highest overtime-attrition correlation.

Recommendation 2 — Conduct a Comprehensive Salary Review

Benchmark current salaries against industry standards, particularly for employees earning below Rs. 5,000/month. Introduce a merit-based salary revision cycle every 6 months for junior-level employees (Job Level 1-2). Focus on Lab Technicians and Sales Representatives, who show the highest attrition among lower-income job roles.

Recommendation 3 — Create a Young Talent Retention Program

Design a structured career development program specifically for employees aged 25-35. This should include mentorship programs, fast-track promotion pathways, skill development allowances, and regular career progression conversations with managers. Employees in this age group are the most mobile and need visible growth opportunities to stay committed.

Recommendation 4 — Introduce Regular Promotion and Recognition Cycles

Employees who have not received a promotion in 2+ years are at significantly higher attrition risk. Implement a structured promotion review every 12-18 months. Beyond formal promotions, introduce non-monetary recognition programs such as Employee of the Quarter, public appreciation, and project ownership opportunities to increase job involvement and commitment.

Recommendation 5 — Strengthen Work-Life Balance Initiatives

Introduce flexible working hours, hybrid work options, and mandatory leave policies. Employees with Work-Life Balance scores of 1-2 should be flagged by HR for immediate one-on-one check-ins. Encourage managers to lead by example — respecting personal time, discouraging after-hours communication, and normalizing mental health days.

Recommendation 6 — Deploy a Predictive Attrition Alert System

Use the trained Random Forest model (88.1% accuracy) to score each employee monthly on their attrition probability. Employees scoring above 70% risk should trigger an automatic HR alert for proactive intervention — such as a career conversation, salary review, or workload assessment. This data-driven early warning system can prevent resignations before they happen.

5. Tools & Technologies Used

This project was built using a combination of Python libraries for data analysis and machine learning, Power BI for interactive visualization, and supporting tools for version control and documentation. Each tool was selected based on its industry relevance and suitability for the specific task.

Tool / Library	Version / Type	Purpose in this Project
Python	3.x	Primary programming language for all data tasks
Pandas	Latest	Data loading, cleaning, filtering and manipulation
NumPy	Latest	Numerical computations and array operations
Matplotlib	Latest	Base plotting library for all chart outputs
Seaborn	Latest	Statistical visualizations — heatmaps, countplots, boxplots
Scikit-learn	Latest	Random Forest model, train-test split, confusion matrix
LabelEncoder	Scikit-learn	Encoding categorical columns for ML model input
Power BI Desktop	2024	Interactive 3-page HR dashboard with KPIs and slicers
DAX	Power BI	Custom measures — Attrition Count, Rate, Avg Income
ReportLab	Python	Programmatic PDF report generation
GitHub	Cloud	Version control, project portfolio and submission
VS Code	Latest	Code editor with Git integration and Jupyter support
Jupyter Notebook	Latest	Interactive coding environment for EDA and ML
Kaggle	Online	Source of IBM HR Analytics Employee Attrition Dataset

All tools used in this project are either open-source or freely available, making this analysis fully reproducible. The complete source code, notebooks, and Power BI dashboard are available on GitHub for reference and review.

6. Steps Involved

The project was executed in six structured phases, each building on the previous to deliver a complete, end-to-end HR analytics solution:

Phase 1 — Dataset & Environment Setup

Downloaded the IBM HR Analytics dataset from Kaggle (1,470 rows, 35 columns). Set up the project folder structure, installed required Python libraries, and verified data integrity. Confirmed zero missing values and identified 4 useless columns (EmployeeCount, EmployeeNumber, Over18, StandardHours) for removal.

Phase 2 — Exploratory Data Analysis (EDA)

Conducted in-depth EDA using Pandas, Matplotlib, and Seaborn. Generated 10 visualizations including attrition by department, age, overtime, income, job satisfaction, and work-life balance. Confirmed 5 out of 5 pre-defined hypotheses through data evidence. All charts saved to outputs folder.

Phase 3 — Machine Learning Model

Built a Random Forest Classifier after encoding categorical variables and splitting data 80:20. Achieved 88.1% accuracy. Generated confusion matrix, classification report, and feature importance chart. Identified Monthly Income, OverTime, and Age as top 3 predictors of attrition.

Phase 4 — Power BI Dashboard

Developed a professional 3-page interactive Power BI dashboard with 4 KPI cards, Department and Gender slicers, navigation buttons, and 8 charts. Pages include Overview, Attrition Analysis, and Employee Details with a Key Insights card and LinkedIn/GitHub profile links.

Phase 5 — Attrition Prevention PDF

Compiled all findings into this structured prevention report using Python ReportLab. Documented 6 data-backed HR recommendations directly targeting the root causes of attrition identified during EDA and Machine Learning phases.

Phase 6 — Final Project Report

Prepared a comprehensive final project report covering Introduction, Abstract, Tools Used, Steps Involved, and Conclusion. All project files pushed to GitHub repository for submission and portfolio display.

7. Conclusion

This HR Analytics project successfully identified the key drivers of employee attrition at IBM and provided a data-driven framework for reducing voluntary turnover. Through rigorous Exploratory Data Analysis and Machine Learning, we established that overtime, low compensation, young employee age, job dissatisfaction, and stagnant career growth are the primary factors driving employees to resign.

The Random Forest model, achieving 88.1% accuracy, demonstrated that employee attrition is highly predictable when the right variables are analyzed. This opens the door to a proactive HR strategy — one where potential resignations are flagged before they occur, and targeted interventions are deployed to retain valuable talent.

The six recommendations outlined in this report — from overtime management to predictive alert systems — are practical, implementable, and directly supported by the data. If adopted, these measures have the potential to significantly reduce the current 16.12% attrition rate, saving the organization both financial resources and human capital.

This project demonstrates how data science, when applied to human resource challenges, can transform reactive HR management into a strategic, evidence-based function that drives organizational success.

Project Summary

Dataset: IBM HR Analytics (1,470 employees, 35 features) | Attrition Rate: 16.12% | ML Model: Random Forest | Accuracy: 88.1% | Top Predictor: Monthly Income | Deliverables: EDA Notebook + ML Model + Power BI Dashboard + Prevention PDF

Future Scope

While this project provides strong predictive and analytical results, several enhancements can be explored in future iterations. These include applying SMOTE (Synthetic Minority Oversampling Technique) to handle the class imbalance problem more effectively, experimenting with XGBoost and Neural Network models for potentially higher accuracy, and building a real-time Streamlit web application that allows HR managers to input employee data and receive instant attrition risk scores.

Additionally, integrating employee exit interview data with quantitative metrics could provide deeper qualitative insights, making the prevention recommendations even more targeted and effective. The ultimate goal is to evolve this project into a fully automated HR intelligence system that continuously monitors workforce health and flags risks in real time.